

COMPLEX NUMBERS ...

SUB TOPICS

⇒ Polar form :

⇒ Euler form :

⇒ Basics of Complex Numbers

⇒ Logarithm of Complex Numbers

⇒ Geometry :

- Circle
- Parabola
- Straight Line
- Point's ...

⇒ Cube roots of Unity

⇒ Binomial Concept with Complex Numbers

⇒ De Moivre's Theorem

Formulas

Complex NO. Basic

Short Notes:

Value of Some trigono:

$$* \cos(\text{even } \pi) = 1$$

$$* \cos(\text{odd } \pi) = -1$$

$$* \sin(\text{even } \pi) = 0$$

$$* \sin(\text{odd } \pi) = 0$$

$$\text{ROOTS OF } x^2 + x + 1 = 0$$

$$\text{ROOTS OF } x^2 - x + 1 = 0$$

$$\omega^2 = \frac{-1 - \sqrt{3}i}{2}$$

$$\omega = \frac{-1 + \sqrt{3}i}{2}$$

$$1 + \omega + \omega^2 = 0$$

$$\text{i. } x > 0, y > 0 \quad \alpha = \tan^{-1}\left(\frac{y}{x}\right)$$

$$\text{ii. } x < 0, y > 0 \quad \alpha = \pi + \tan^{-1}\left(\frac{y}{x}\right)$$

$$\text{iii. } x < 0, y < 0 \quad \alpha = -\pi + \tan^{-1}\left(\frac{y}{x}\right)$$

$$\text{iv. } x > 0, y < 0 \quad \alpha = -\tan^{-1}\left(\frac{y}{x}\right)$$

Logarithm Of Complex Numbers

Argument $\hat{=}$ Amplitude

$$\text{i. } \text{Arg}(x+iy) = \tan^{-1}\left(\frac{\text{Im}(z)}{\text{Re}(z)}\right) \Rightarrow \text{Arg}(z) = -\text{Arg}(\bar{z})$$

$$\text{ii. } |z| = |x+iy| = \sqrt{x^2 + y^2} \Rightarrow \text{Arg}(z^n) = n \text{Arg}(z)$$

$$\text{iii. } \cos \theta + i \sin \theta = \text{cis } \theta \Rightarrow \text{Arg}(z_1 z_2) = \text{Arg}(z_1) + \text{Arg}(z_2) + 2k\pi$$

$$\text{iv. } \boxed{\begin{aligned} |\text{cis } \theta| &= 1 \\ |\text{cis}(-\theta)| &= 1 \end{aligned}} \Rightarrow \text{Arg}\left(\frac{z_1}{z_2}\right) = \text{Arg}(z_1) - \text{Arg}(z_2)$$

$$\text{v. } z = |z| \text{cis}(\theta) \quad (\because |\bar{z}| = |z|) \Rightarrow \text{Arg}\left(\frac{z_1}{z_2}\right) = 2k\pi - \theta$$

$$\text{vi. } \text{Arg}(\bar{z}) = -\theta \quad \text{If } z = -2 \quad \text{Arg } z = \pi$$

$$\text{Arg}(\bar{z}) = -\theta \quad \bar{z} = -2 \quad \text{Arg } \bar{z} = \pi$$

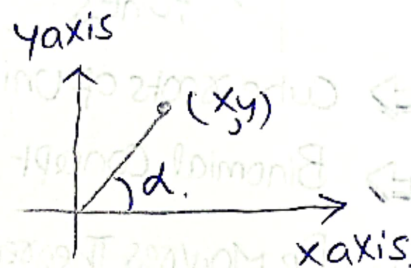
$$\text{vii. } |z| = |\bar{z}|$$

- Addition
- Subtraction
- Multiplication/Division (Rational)
- Square root ...

$$\text{i. } \omega^3 = 1$$

$$\text{ii. } \omega^{3n+1} = \omega$$

$$\text{iii. } \omega^{3n+2} = \omega^2$$



Properties of cis:

$$i. \text{cis } \alpha \text{cis } \beta = e^{i\alpha} e^{i\beta} = e^{i(\alpha+\beta)} = \text{cis}(\alpha+\beta)$$

$$ii. \text{cis } \alpha \text{cis } \beta \text{cis } \gamma = \text{cis}(\alpha+\beta+\gamma)$$

$$iii. \frac{\text{cis } \alpha}{\text{cis } \beta} = \text{cis}(\alpha-\beta)$$

$$iv. (\text{cis } \alpha)^n = \text{cis } n\alpha$$

$$\boxed{\text{cis } \theta = \cos \theta + i \sin \theta = e^{i\theta}}$$

$$v. \frac{1}{\text{cis } \theta} = \frac{1}{e^{i\theta}} = e^{-i\theta} = e^{i(-\theta)} = \text{cis}(-\theta)$$

$$\boxed{i^i = e^{-\pi/2}}$$

$|z| = \text{constant}$ Always represents Circle.

Inequality of Complex Numbers:

$$||z_1| - |z_2|| \leq |z_1 + z_2| \leq |z_1| + |z_2|$$

$$||z_1| - |z_2|| \leq |z_1 - z_2| \leq |z_1| + |z_2|$$

Conversion of Euler form.

For $z = x + iy$

$$z = (\sqrt{x^2 + y^2}) \left(\frac{x}{\sqrt{x^2 + y^2}} + i \frac{y}{\sqrt{x^2 + y^2}} \right)$$

n^{th} roots of Unity:

Let $1, \alpha_1, \alpha_2, \dots, \alpha_{n-1}$ are n^{th} roots of Unity then

$$z^n - 1 = (z - 1)(z - \alpha_1)(z - \alpha_2) \dots (z - \alpha_{n-1})$$

$$z^n - 1 = (1 - z)(\alpha_1 - z)(\alpha_2 - z) \dots (\alpha_{n-1} - z)$$

Imag Part

Complex NO. Plan.

Real Part

De Moivre's Theorem

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

$$i. (\cos \theta + i \sin \theta)^n \neq \cos n\theta + i \sin n\theta$$

$$(\sin \theta \pm i \cos \theta)^n \neq \sin n\theta \pm i \cos n\theta$$

$$ii. (\cos \theta + i \sin \theta)^{\frac{p}{q}} = (\cos \frac{p}{q}\theta + i \sin \frac{p}{q}\theta)$$

$$iii. \frac{1}{\cos \theta + i \sin \theta} = \cos \theta - i \sin \theta$$

(i) n^{th} roots of Unity lie on Unit circle $|z|=1$ and
Divide its circumference into n equal parts

$$z = \cos \frac{2k\pi}{n} + i \sin \frac{2k\pi}{n}$$

$$|z_1 + z_2| = a : \text{Circle}$$

$$|z_1 + z_2| = 2a : \text{Ellipses}$$

$$|z_1 + z_2 + z_3| = a : \text{Sphere}$$

$$|z_1 - z_2| = 2a : \text{Hyperbola}$$

$$(ii) i + i^2 + i^3 + i^4 = 0$$